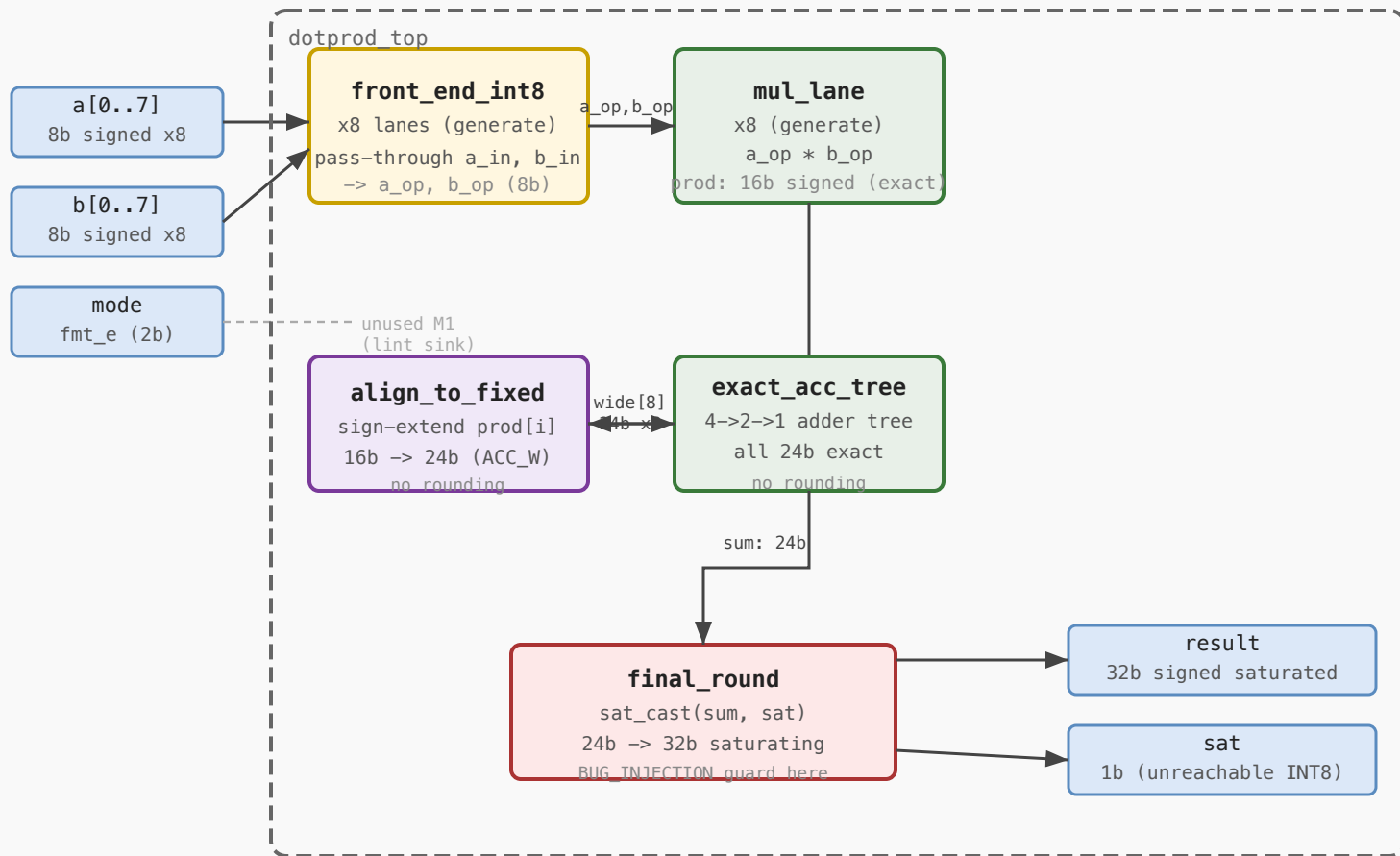


dotprod_top – INT8 Datapath (M1)

N_LANES=8, INT8_W=8, PROD_W=16, ACC_W=24, INT8_OUT_W=32



Legend: front-end / decode multiply / accumulate alignment / extension rounding / saturation

All paths combinational (no clock/reset). Saturation unreachable for INT8 single-shot (max |sum| = 131072, needs 19b; INT8_OUT_W = 32b).